

Decentralized Spatial Negotiation via Residual Utility Learning

Learning to Negotiate Chaos: Residual Multi-Agent Policies over Behavioral Priors

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Abstract—Traffic in many developing regions operates under weak lane discipline, heterogeneous vehicle interactions, and decentralized negotiation of shared space. These environments challenge conventional lane-based trajectory prediction and motion planning frameworks, which typically rely on structured map priors and predefined leader-follower relationships. This paper presents a utility-guided residual multi-agent reinforcement learning framework for motion modeling in weakly structured non-lane-based traffic.

The proposed approach combines an interpretable utility-based behavioral model with a residual multi-agent reinforcement learning policy. The utility model encodes directional preference, speed incentive, collision risk, and proximity-aware interaction terms derived from behavioral and economic decision-making principles. Rather than learning motion policies from scratch, the reinforcement learning layer adaptively modulates the utility-driven behavior through residual policy learning, enabling emergent negotiation and interaction behaviors while preserving physical interpretability and safety consistency.

The framework is evaluated using high-resolution aerial trajectory data collected from heterogeneous traffic environments in Lebanon, including a weak-lane multilane freeway and a chaotic urban roundabout. The proposed method is benchmarked against deterministic interaction models (ORCA), stochastic utility-based models, self-driven particle formulations, and purely learning-based reinforcement learning approaches without behavioral priors. Results demonstrate that the proposed utility-guided residual learning framework achieves improved trajectory realism, interaction stability, and behavioral fidelity under dense non-lane-based conditions while maintaining interpretable decision structures.

This work establishes a principled framework for integrating behavioral utility theory and multi-agent reinforcement learning for decentralized motion generation in weakly structured traffic environments.

Index Terms—Motion Modeling, Reinforcement Learning, Multi-Agent Systems, Decision-Making Under Uncertainty, Trajectory Data, Model Calibration

I. INTRODUCTION

In many developing countries, transportation systems operate under conditions that differ substantially from the structured, lane-disciplined traffic typically assumed in classical models [1]–[3]. Weak rule enforcement, heterogeneous vehicle

fleets, and informal negotiation of right-of-way produce complex interactions that challenge traditional traffic flow theory [4]. Such irregular behaviors including lateral encroachments, opportunistic yielding, and abrupt accelerations are prevalent in Lebanon, where infrastructure and enforcement limitations contribute to weak-lane and non-standard driving patterns [5]. Accurately representing these settings requires empirical data and modeling approaches that remain valid when lane structure and deterministic leader-follower assumptions are only weakly satisfied.

Recent advances in aerial data collection enable direct observation of naturalistic traffic with high spatial and temporal resolution. Drone-based video provides an unobstructed top-down view of vehicle motion, allowing large-scale extraction of trajectories while preserving behavioral authenticity [6]. This study consolidates two complementary aerial datasets collected in Jounieh, Lebanon: (i) a multilane freeway segment characterized by non-lane-based longitudinal interactions and informal lateral negotiation, and (ii) a nominally single-lane roundabout exhibiting opportunistic merging and chaotic circulation. These sites differ in geometric constraints, conflict resolution mechanisms, and driver cooperation, providing a rare empirical basis for benchmarking multiple modeling paradigms under weak-lane conditions.

Microscopic vehicle models and modern trajectory forecasting frameworks remain central to representing vehicle interactions. However, state-of-the-art deep learning architectures (e.g., LaneGCN, VectorNet, Wayformer) predominantly assume strict lane centerlines, predefined map topologies, and structured drivable spaces. Consequently, they struggle to capture decentralized negotiation, emergent lane formation, and interaction-only forecasting without map priors in weak-lane environments. Furthermore, while Multi-Agent Reinforcement Learning (MARL) offers a pathway to game-theoretic trajectory generation, standard MARL formulations frequently fail in traffic applications. These failures stem from enormous action spaces, sparse rewards, collision instability, and a lack of human behavioral priors, ultimately yielding physically unrealistic driving behavior.

To bridge this gap, we introduce a utility-guided residual MARL architecture specifically designed for unstructured, non-lane-based traffic. By decoupling human behavioral priors from deep policy learning, the proposed framework avoids the pitfalls of blind reinforcement learning. The architecture operates through a four-stage process:

- **Stage 1: Utility Prior:** An interpretable baseline utility formulation (U_{base}) captures directional preference, collision risk, and speed incentive to generate candidate trajectories and nominal actions based on established behavioral theories (e.g., Prospect Theory).
- **Stage 2: Interaction Graph:** A local interaction graph is constructed to capture relative states, time-to-collision (TTC), proximity, and velocity differences, utilizing attention mechanisms to process heterogeneous neighborhood data.
- **Stage 3: RL Residual:** Instead of learning raw control actions, the MARL policy learns a residual behavioral adjustment (ΔU or $\Delta \Theta$) to modulate the underlying utility parameters.
- **Stage 4: Final Utility Maximization:** The final action is determined by maximizing the combined utility ($U_{final} = U_{base} + \lambda \Delta U$), ensuring that the output preserves physical realism, interpretability, and safety controllability while enabling emergent social behaviors.

Beyond reproducing vehicle kinematics, weak-lane and chaotic driving reflect asymmetric human decision-making under perceived risk. Prospect theory provides a principled framework to quantify such behaviors relative to reference points, highlighting how drivers trade off progress against conflict risk. The proposed residual MARL framework complements this by providing a data-driven mechanism to capture implicit interaction rules and spatial negotiation across multiple agents in dense settings.

The contributions of this study are:

- Provision of a consolidated aerial trajectory dataset for two weak-lane traffic regimes in Lebanon.
- Development of a novel utility-guided residual MARL framework that enables interaction-only forecasting and decentralized negotiation without map priors.
- Calibration and benchmarking of deterministic interaction models (ORCA), stochastic utility formulations (PT), and the proposed hybrid MARL under consistent procedures across both sites.
- Behavioral interpretation linking observed aggressiveness to stochastic utility parameters, demonstrating MARL’s capability to encode complex, emergent traffic interactions while maintaining physical interpretability.

This integrated approach establishes a robust foundation for modeling heterogeneous, non-lane-based traffic using a principled combination of deterministic, stochastic, and learning-based methods.

II. LITERATURE REVIEW

Modeling vehicle motion in traffic systems has been a central topic in transportation research, with a wide range

of approaches developed to describe how drivers interact with their surroundings. Traditional models largely rely on structured, lane-based assumptions, while more recent work has explored interaction-driven, behavioral, and learning-based frameworks to better capture complex and unstructured traffic environments. This section reviews the relevant literature across four key domains: classical microscopic models, non-lane-based traffic modeling, interaction-based motion models, and data-driven learning approaches.

A. Classical Microscopic Traffic Models

Microscopic traffic flow models describe the behavior of individual vehicles, typically focusing on how drivers adjust acceleration in response to surrounding traffic conditions. Among the most widely used formulations are car-following models such as the Intelligent Driver Model (IDM) [7], [8], the Full Velocity Difference Model (FVDM) [9], and Gipps’ model. These models assume a well-defined leader–follower relationship and describe vehicle acceleration as a function of spacing, relative speed, and desired velocity.

Such formulations have been successful in modeling traffic dynamics on structured highways with strong lane discipline. Their parameters are often interpretable, capturing driver preferences for desired speed, time headway, and acceleration behavior. However, their reliance on a single leading vehicle and strictly longitudinal interactions limits their applicability in environments where lane structure is weak or absent. In such cases, interactions become inherently two-dimensional, and drivers may respond simultaneously to multiple neighboring vehicles rather than a single leader.

B. Non-Lane-Based and Heterogeneous Traffic

In many developing regions, traffic is characterized by weak lane discipline, heterogeneous vehicle types, and informal negotiation of space. These conditions give rise to non-lane-based traffic, where vehicles do not adhere to fixed lanes and instead move in a continuous two-dimensional space. Several studies have highlighted the limitations of classical car-following models in such environments and proposed adaptations to account for lateral interactions and multiple leaders.

Research on heterogeneous traffic flow emphasizes the importance of capturing interactions beyond a single leader, including lateral encroachments and opportunistic maneuvers. Approaches often rely on data-driven definitions of interaction neighborhoods or extend classical models to consider multiple surrounding vehicles. Despite these efforts, many models retain a fundamentally longitudinal structure and struggle to fully represent the decentralized and multi-agent nature of non-lane-based traffic. This has motivated the exploration of alternative modeling paradigms that operate directly in continuous space.

C. Interaction-Based and Continuous-Space Models

To address the limitations of lane-based formulations, a class of models has emerged that represents vehicle motion

through local interactions in continuous space. One prominent example is the Social Force Model (SFM) [10], originally developed for pedestrian dynamics. In this framework, agents are subject to attractive forces toward desired motion and repulsive forces from nearby agents, enabling flexible modeling of dense and unstructured interactions.

In robotics and multi-agent systems, collision avoidance has been extensively studied using velocity-based approaches such as Velocity Obstacles (VO) and Reciprocal Velocity Obstacles (RVO). Building on these concepts, the Optimal Reciprocal Collision Avoidance (ORCA) framework [?], [?] formulates collision avoidance as a set of linear constraints in velocity space. Each agent computes a feasible set of collision-free velocities by considering pairwise interactions with neighboring agents and selects the velocity closest to its preferred motion.

These approaches are well-suited for modeling multi-agent interactions in continuous space and can scale to dense environments without requiring explicit lane definitions. However, they are primarily geometric and deterministic in nature, focusing on collision avoidance rather than behavioral realism. As a result, they do not explicitly capture driver heterogeneity, risk perception, or decision-making under uncertainty, which are critical components of real-world driving behavior.

D. Behavioral and Stochastic Models

Human driving behavior is inherently stochastic and influenced by subjective perceptions of risk and reward. Behavioral models aim to capture these aspects by incorporating psychological and decision-theoretic principles into motion modeling. Prospect Theory [?] provides a well-established framework for modeling decision-making under risk, where outcomes are evaluated relative to a reference point and losses are weighted more heavily than gains.

In transportation research, prospect-theoretic formulations have been applied to problems such as route choice and gap acceptance, demonstrating the importance of asymmetric risk perception in driving decisions. These models provide a natural way to represent aggressive or conservative behavior, particularly in environments where drivers must continuously evaluate trade-offs between progress and safety.

Despite their strengths, most behavioral models are not formulated in a fully continuous two-dimensional setting and often rely on simplified interaction assumptions. Integrating behavioral decision-making with realistic multi-agent interactions remains an open challenge, particularly in non-lane-based traffic environments.

E. Learning-Based and Multi-Agent Approaches

Recent advances in machine learning have led to the development of data-driven approaches for modeling vehicle behavior, particularly through reinforcement learning. In this paradigm, agents learn control policies by interacting with an environment and optimizing a reward function that encodes desired objectives such as safety, efficiency, and comfort.

Multi-Agent Reinforcement Learning (MARL) extends this framework to settings where multiple agents interact simultaneously. The Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm [?] is a widely used approach for continuous control problems, enabling agents to learn decentralized policies while leveraging centralized training. Such methods have been applied to autonomous driving scenarios, demonstrating the ability to capture complex interaction patterns and emergent behaviors.

Learning-based approaches offer significant flexibility and can adapt to complex, high-dimensional environments without requiring explicit modeling assumptions. However, they often lack interpretability and depend heavily on the design of reward functions and training data. Moreover, their behavior may be difficult to analyze or generalize beyond the training conditions.

F. Research Gap and Contribution

The existing literature provides a diverse set of tools for modeling traffic dynamics, ranging from deterministic interaction-based models to stochastic behavioral formulations and data-driven learning approaches. However, these paradigms are typically studied in isolation, and direct comparisons across fundamentally different modeling assumptions remain limited.

In particular, there is a lack of unified frameworks that evaluate deterministic collision-avoidance models, stochastic decision-theoretic models, and multi-agent learning approaches within the same non-lane-based traffic context. This gap is especially relevant for heterogeneous traffic environments, where interactions are multi-dimensional and behavior is highly variable.

To address this limitation, the present study develops a comparative framework that integrates three complementary approaches: ORCA as a deterministic interaction model, a prospect-theoretic formulation as a stochastic behavioral model, and MADDPG as a learning-based multi-agent model. By calibrating and evaluating these models on a common dataset of non-lane-based traffic, this work provides new insights into their relative strengths and limitations in capturing complex driving behavior.

G. Aerial Trajectory Data and Roundabout Analysis

The rapid growth of aerial data collection has transformed how chaotic and non-lane-based traffic patterns are observed and modeled. Unmanned aerial vehicles (UAVs) provide a top-down view that can simultaneously capture trajectories of all road users, including vehicles, cyclists, and pedestrians, over complex geometries. A UAV-based methodological framework was developed for multivehicle trajectory extraction at urban roundabouts, demonstrating that drones can capture detailed kinematic profiles and gap-acceptance behavior that are difficult to observe using fixed cameras [11]. Other researchers used drones to analyze traffic flow at urban roundabouts, showing how high-resolution aerial imagery can reveal subtle lane

encroachments, yielding patterns, and circulation strategies that are often invisible in aggregated loop-detector data [12].

In parallel, several open datasets have emerged that provide high-fidelity trajectories from aerial views for intersections and roundabouts. Researchers introduced the TUMDOT-MUC dataset, a large-scale collection of multimodal trajectories (motorized vehicles and vulnerable road users) gathered by drone over multiple urban sites, specifically designed to support model calibration and safety analysis in complex multimodal environments [13]. Other researchers published a dataset of UAV-based roundabout aerial images with detailed vehicle annotations, enabling researchers to build and validate computer-vision pipelines for vehicle detection and, by extension, trajectory extraction in circular intersections [14]. More recently, the MiTra dataset was released, which provides drone-based trajectories for freeways with ramps across all traffic states, illustrating how persistent aerial observation can be used to study microscopic behavior from free flow to congestion [15].

Aerial data has also been leveraged for proactive safety analysis. The Extra Brake Required to Avoid a Crash (EBRAC) surrogate measure was proposed using drone-based image-recording data, quantifying near-crash conflicts at intersections by reconstructing detailed braking profiles and interaction dynamics [16]. This work illustrates how high-resolution trajectories can be transformed into interpretable safety indicators that directly reflect road-user maneuvers. Similar methodologies have been applied to other drone-based datasets to derive conflict-based safety metrics and evaluate how geometric design and control strategies affect risk.

H. Research Gap

Overall, the literature shows (i) strong evidence that weak lane discipline and heterogeneous traffic can lead to highly irregular, “chaotic” driving patterns; (ii) a growing family of non-lane-based car-following and agent-based models tailored to disordered streams; and (iii) an expanding ecosystem of aerial trajectory datasets that enable detailed observation of such behavior. However, gaps remain in how these methods are applied across different geometric contexts.

For freeway environments in developing countries, only limited work has calibrated microscopic models to non-lane-based longitudinal behavior, where drivers frequently deviate from lane centers, negotiate lateral space informally, and adopt highly variable spacing at moderate to high speeds. Existing studies rarely evaluate how classical lane-based models such as IDM perform relative to non-lane-based formulations when exposed to such freeway conditions.

Similarly, for small, nominally single-lane roundabouts, prior research rarely calibrates or compares models using high-frequency trajectories that capture the lane-free circulation, opportunistic merging, and lateral encroachment characteristic of these sites. In particular, there is little evidence on whether IDM can adapt to these chaotic interaction patterns or whether non-lane-based models (e.g., FVDM, two-sided gap models, SFM) offer clear advantages.

The present study addresses these gaps by (i) extracting detailed vehicle trajectories from aerial video at both freeway and roundabout sites in Lebanon, (ii) calibrating lane-based (IDM) and non-lane-based models (FVDM and SFM) to these contrasting environments, and (iii) systematically comparing their ability to reproduce the observed acceleration, spacing, and maneuvering patterns. This dual-context evaluation enables a unified assessment of model robustness across two representative forms of non-lane-based driving: high-speed freeway flow and chaotic roundabout circulation.

III. METHODOLOGY

This work proposes a utility-guided residual multi-agent reinforcement learning framework for motion generation in weakly structured non-lane-based traffic. Unlike conventional lane-based trajectory prediction approaches that rely on explicit lane topology, map priors, or predefined leader-follower relationships, the proposed framework models vehicle interactions directly in continuous two-dimensional space through decentralized utility maximization and adaptive behavioral learning.

The proposed framework combines two complementary components: 1) an interpretable utility-based behavioral prior that captures directional preference, collision risk, speed incentive, and local interaction behavior; and 2) a residual multi-agent reinforcement learning policy that adaptively modulates behavioral utility parameters according to the surrounding traffic context.

Rather than directly learning low-level control actions or trajectories from scratch, the reinforcement learning layer learns residual behavioral adaptations over a structured utility formulation. This preserves physical interpretability and stability while enabling emergent negotiation behaviors in dense heterogeneous traffic.

A. Problem Formulation

Traffic is modeled as a decentralized multi-agent system operating in continuous two-dimensional space. Let $\mathcal{A} = \{1, 2, \dots, N\}$ denote the set of traffic agents. Each agent $i \in \mathcal{A}$ is represented by a continuous state:

$$s_i^t = [x_i^t, y_i^t, v_i^t, \theta_i^t] \quad (1)$$

where (x_i^t, y_i^t) denotes the spatial position, v_i^t denotes the speed magnitude, and θ_i^t denotes the heading angle. Each agent observes a local neighborhood:

$$o_i^t = [s_i^t, \{s_j^t\}_{j \in \mathcal{N}_i}] \quad (2)$$

where $\mathcal{N}_i$ denotes neighboring agents within a perception radius R_p .

The objective of each agent is to generate motion trajectories that progress toward the desired destination, avoid collisions, maintain smooth and feasible motion, and negotiate shared space with neighboring agents. Unlike lane-based formulations, no predefined lane centerlines or map topology constraints are imposed.

B. Utility-Based Behavioral Prior

The nominal motion behavior of each agent is governed by a utility-maximization process. At each time step, the agent selects the candidate speed and heading pair that maximizes its expected utility:

$$(v_i^*, \theta_i^*) = \arg \max_{v, \theta} U_i(v, \theta; \Theta_i) \quad (3)$$

where Θ_i denotes the behavioral utility parameters controlling the agent's driving characteristics. The total utility function consists of five primary components:

$$U_i = U_{dir} + U_{speed} + U_{dist} - U_{col} - U_{path} \quad (4)$$

1) *Directional Utility*: The directional utility encourages motion toward the desired destination direction:

$$U_{dir} = S_\theta \cos(\theta - \theta_{goal}) \quad (5)$$

where θ is the candidate heading, θ_{goal} is the desired destination heading, and S_θ controls directional sensitivity.

2) *Speed Incentive*: To allow dynamic adaptation to pacing vehicles in dense traffic, the speed utility measures how the candidate reference speed compares to the ego agent's current speed. Let $\rho_g = \frac{v_{ref}}{v_{current}}$. The speed incentive is defined as:

$$U_{speed} = S_v \frac{\rho_g}{1 + \rho_g^{(\xi_i - 1)/2}} \quad (6)$$

where S_v controls the reward magnitude and $\xi_i > 1$ modulates how sharply the utility drops as the speed deviates from the reference, reflecting driver discomfort with speed mismatches.

3) *Proximity Incentive*: To encourage progress toward the destination, a horizon-normalized proximity reward is applied rather than a simple geometric repulsion:

$$U_{dist} = \frac{S_d}{1 + (d_{eff}/H_p)^\gamma} \quad (7)$$

where d_{eff} is the effective distance to the destination, and $H_p = \kappa v_i$ is a dynamic perception horizon that scales with speed. The parameter S_d caps the term, and γ controls how steeply the reward decreases as the distance exceeds the perception horizon.

4) *Collision-Risk Utility*: In highly unstructured, non-lane-based environments, collision risk cannot be reduced to simple bounding-box overlaps or static distance thresholds. Instead, it is modeled probabilistically to capture the inherent spatial and temporal uncertainty of surrounding agents' trajectories. This sophisticated presence model evaluates the precise likelihood of the ego agent i and any neighboring agent j simultaneously occupying a candidate spatial point g .

First, we determine the anticipated arrival time $\Delta t_{i,g}$ for the ego agent to reach the candidate point g , assuming a constant current velocity:

$$\Delta t_{i,g} = \frac{\|\vec{r}_g - \vec{r}_i\|}{\|\vec{v}_i\| + \epsilon} \quad (8)$$

where $\vec{r}_g$ is the coordinate of the candidate point, $\vec{r}_i$ and $\vec{v}_i$ are the current position and velocity vectors of the ego agent, and ϵ is a small constant to prevent division by zero.

Next, the future state of each neighboring agent $j \in \mathcal{N}_i$ is projected forward to time $t + \Delta t_{i,g}$. Because agent behavior is stochastic, this future position is modeled as a multivariate normal distribution $\vec{r}_j(t + \Delta t_{i,g}) \sim \mathcal{N}(\vec{\mu}_j, \Sigma_j)$, where the mean predicted position $\vec{\mu}_j$ is defined by linear kinematic extrapolation:

$$\vec{\mu}_j = \vec{r}_j(t) + \vec{v}_j \Delta t_{i,g} \quad (9)$$

and Σ_j is the covariance matrix representing the prediction uncertainty across the spatial dimensions (e.g., $\Sigma_j = \text{diag}(\sigma_x^2, \sigma_y^2)$).

The specific collision probability $P_{j,g}$, representing the likelihood of agent j occupying point g at the exact moment agent i arrives, is computed using the probability density function of this multivariate normal distribution:

$$P_{j,g} \approx \frac{1}{(2\pi)^{d/2} |\Sigma_j|^{1/2}} \exp \left(-\frac{1}{2} (\vec{r}_g - \vec{\mu}_j)^T \Sigma_j^{-1} (\vec{r}_g - \vec{\mu}_j) \right) \quad (10)$$

where d is the number of spatial dimensions (e.g., $d = 2$ for ground traffic). The term inside the exponential represents the squared Mahalanobis distance, which normalizes the physical distance between the candidate point and the neighbor's predicted position according to the directional uncertainty of the traffic flow.

To evaluate the total collision risk for the candidate point g , the individual presence probabilities of all agents within the perception horizon are aggregated:

$$P_{i,g} = \sum_{j \in \mathcal{N}_i} P_{j,g} \quad (11)$$

Finally, this cumulative probability is scaled into a utility penalty. This continuous, differentiable risk landscape provides a critical gradient for the residual reinforcement learning policy to navigate safely:

$$U_{col} = W_{cc} \cdot P_{i,g} \quad (12)$$

where W_{cc} is the severity weight parameter that controls the agent's absolute risk aversion.

5) *Path Adherence and Boundary Constraints*: While the agents operate in a decentralized, non-lane-based environment, they are still constrained by the physical geometry of the drivable space (e.g., highway edges or roundabout outer boundaries). To enforce these infrastructural limits without imposing artificial lane centerlines, a boundary adherence penalty is introduced:

$$U_{path} = w_p (1 - e^{-\beta l_i^2}) \quad (13)$$

where l_i represents the lateral deviation or proximity to the absolute infrastructural boundary of the road network, w_p scales the maximum penalty for driving off-road, and β adjusts how sharply the penalty increases as the vehicle approaches the edge. This formulation guarantees that agents freely negotiate lateral space within the roadway while learning to avoid infrastructural boundaries.

C. Residual Multi-Agent Reinforcement Learning

Fixed utility parameters cannot fully capture adaptive negotiation strategies, contextual aggressiveness, cooperative yielding, or emergent interaction behaviors observed in weakly structured traffic. Therefore, a residual multi-agent reinforcement learning (MARL) policy adaptively modulates the utility parameters according to the surrounding traffic context.

The residual policy is defined as:

$$\Delta\Theta_i^t = \pi_\phi(o_i^t) \quad (14)$$

where π_ϕ denotes the learned policy with neural network parameters ϕ , and $\Delta\Theta_i^t$ represents the utility parameter residuals.

The updated adaptive utility parameters become:

$$\Theta_i^t = \Theta_i^{base} + \Delta\Theta_i^t \quad (15)$$

The motion is then evaluated using the updated utility:

$$(v_i^*, \theta_i^*) = \arg \max_{v, \theta} U_i(v, \theta; \Theta_i^t) \quad (16)$$

D. Policy Observation Space

Each agent observes entirely local and decentralized traffic states:

$$o_i^t = [x_i, y_i, v_i, \theta_i, \theta_{goal}, \{\Delta x_{ij}, \Delta y_{ij}, \Delta v_{ij}\}_{j \in \mathcal{N}_i}] \quad (17)$$

where Δx_{ij} , Δy_{ij} , and Δv_{ij} denote relative positions and velocity differences with neighbors.

E. Residual Action Space

To fully leverage the behavioral structure, the RL policy modulates both the magnitude (weights) and the shape (sensitivities) of the utility components. The continuous action space is defined as:

$$\Delta\Theta_i^t = [\Delta S_v, \Delta S_\theta, \Delta S_d, \Delta W_{cc}, \Delta \xi_i, \Delta \gamma, \Delta w_p] \quad (18)$$

This allows the policy to dynamically adapt the agent's aggressiveness, directional commitment, collision sensitivity, speed mismatch tolerance (ξ_i), horizon focus (γ), and boundary adherence (w_p).

F. Vehicle Motion Update

After selecting the optimal speed and heading pair (v_i^*, θ_i^*) , deterministic accelerations are computed:

$$a_x = \frac{v_i^* \cos \theta_i^* - v_{x,i}}{\Delta t}, \quad a_y = \frac{v_i^* \sin \theta_i^* - v_{y,i}}{\Delta t} \quad (19)$$

Vehicle states are then updated forward in time via discrete kinematic integration.

G. Reward Function

The MARL policy is trained using a multi-objective reward balancing progress, safety, smoothness, and trajectory realism:

$$R_i = w_1 R_{progress} + w_2 R_{safety} + w_3 R_{smooth} + w_4 R_{traj} \quad (20)$$

where:

$$R_{progress} = v_i \cos(\theta_i - \theta_{goal}) \quad (21)$$

$$R_{safety} = - \sum_{j \in \mathcal{N}_i} \exp(-d_{ij}) \quad (22)$$

$$R_{smooth} = -(a_x^2 + a_y^2) \quad (23)$$

$$R_{traj} = -||x_i^{pred} - x_i^{gt}||_2 \quad (24)$$

H. Training Procedure

The residual policy is trained using centralized training and decentralized execution. Training episodes are initialized using observed traffic states extracted from the drone-based trajectory data. Importantly, the policy does not directly control steering or acceleration; vehicle motion remains governed by the modulated utility maximization and kinematic constraints, preserving interpretability while enabling complex emergent interactions.

IV. DATA COLLECTION AND TRAJECTORY EXTRACTION

V. RESULTS

Add data extraction results to previous section
Results and Discussion:
Add comparison section for roundabouts
Add comparison section for highway
Add PT part

VI. CONCLUSION

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